Hossein Aqasizade

Website | GitHub
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Graduate Research Assistant, Department of Computer Engineering,
University of Mazandaran, Babolsar, Iran

Educational Background

Master Degree (2020 - 2023)

M.Sc., Computer Engineering, Computer Networks
Department of Computer Engineering, University of Mazandaran, Babolsar, Iran
Thesis: Combining Virtual Machines and Containers in a Cluster
Supervised by: Dr. Ehsan Ataie
Total GPA: 3.88/4 (17.92/20 in Iranian Scale)

Bachelor Degree (2016-2020)

B.Sc., Computer Engineering, Information Technology
Department of Computer Engineering, Azad University, Mazandaran, Iran
Final Project: Implemented management system with virtualization
Advised by: Dr. Esmaeili
Total GPA: 3/4 (14.49/20 in Iranian Scale)

Academic Experiences

Teaching Assistant

- Cloud Computing (for Master Students), Supervised by Dr. Ehsan Ataie, Fall 2022
  Training (Kubernetes, Docker, Prometheus and Grafana), and Project Management
- Cloud Computing (for Bachelor Students), Supervised by Dr. Ehsan Ataie, Fall 2022
  Training (Kubernetes, Docker, Prometheus and Grafana) and Project Management
- Performance Evaluation (for Master Students), Supervised by Dr. Ehsan Ataie, Fall 2021
  Training, Exercise Definition, and Project Management

Research Assistant

- Performance Analysis of Serverless Platforms, Supervised by Dr. Ehsan Ataie, (Nov 2022 - Aug 2023)
  As a team leader, my work revolves around exploring serverless platforms such as Open-FaaS, Fission, Nuclio, Knative, and OpenWhisk. The primary objective of our project is to assess the performance of these platforms in different Kubernetes deployments under varying conditions.

Journal Papers

Performance Analysis of Containers Running on Top of Virtual Machines (in preparation)

Description: Investigating the performance implications of running containers within virtualized environments. This study aims to quantify resource overhead and identify optimization strategies for containerized applications.
Skills Developed: Proficiency in containerization, virtualization, and performance testing methodologies.
Journal: Cluster Computing

Optimizing Container Performance in Kubernetes: A Study with OpenFaaS on Virtual Machines (in preparation)

Description: Studying the performance of serverless functions with OpenFaaS and MySQL on Kubernetes in virtualized environments, including Xen VMs in a CloudLab
Skills Developed: Proficient in Kubernetes, Xen virtualization, OpenFaaS, MySQL, monitoring tools and benchmarks.

A Comparative Study of Virtualization: Heavyweight Virtualization or Lightweight Virtualization? (Submitted)

Description: Conducting a comparative analysis to evaluate the performance and resource utilization of heavyweight virtualization and lightweight virtualization techniques.
Skills Developed: Proficiency in virtualization, experimental design, data analysis, and academic writing.
Journal: The CSI Journal on Computing Science and Information Technology

- Performance Evaluation of Kubernetes Deployments Approaches with OpenFaaS Integration (in preparation)

  Description: Investigating the performance implications of integrating OpenFaaS with different deployment approaches in Kubernetes clusters. The study aims to assess resource utilization and scalability of serverless functions within different orchestration frameworks.
  Skills Developed: Proficiency in Kubernetes orchestration, containerization technologies, serverless computing with OpenFaaS, experimental design, and data analysis.

Research Interests

- Serverless Computing
- Machine Learning
- Cloud Computing
- High Performance Computing
- Parallel Computing
- Virtualization

Software Skills

- Kubernetes (show certificate)
- Docker (show certificate)
- LPIC 1 and LPIC 2
- LPIC 304 (show certificate)
- Python Advanced (show certificate)
- AWS ECS (show certificate)
- CloudLab
- NetConf, NetFlow, NetSNMP, NetXMS
- Monitoring Tools (Prometheus, Grafana and Zabbix)
- Microsoft Certificate (70-698, 70-410 and 70-411)
- Benchmarking Tools
- HPC (High Performance Computing)
- Virtualization (Xen, ESXi, KVM and ...)
- CompTIA Network+
- Serverless Platforms (OpenFaaS, Knative, Fission, Nuclio)
- HTML, CSS, JS

Educational Projects

- Implemented **Genetic Algorithm** for feature selection in Mobile Price classification using **Python**.
- Implemented a **Neural Network** for Dry Bean classification using **Python**.
- Configured **NetSNMP** Configuration for Network Monitoring.
- Configured **NetConf** Configuration for Network Device Management.
- Configured **NetFlow** Configuration for Traffic Analysis.
- Configured **NetXMS** Configuration for Network Monitoring.
- Implemented and simulated cloud computing environments using **CloudSim** toolkit in **Eclipse** IDE.
- Designed and deployed a queuing system simulation **(M/M/1/K)** using **Python**.
- Developed a Python program for a **graph-based access control system** using **Python**.

Professional References

- Dr. Ehsan Ataie (Supervisor)

  Assistant Professor, Department of Computer Engineering,
  University of Mazandaran, Babolsar, Iran

- Dr. Mostafa Bastam (Advisor)

  Assistant Professor, Department of Computer Engineering,
  University of Mazandaran, Babolsar, Iran

- Dr. Payam Mahmoudi-Nasr

  Assistant Professor, Department of Computer Engineering,
  University of Mazandaran, Babolsar, Iran